Quantum Gravitational Dynamics

Chapter 3: Field Equations

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Abstract

We derive the dynamical equations governing the graviton field σ_μ from the QGD action. Variation with respect to σ_μ yields a fourth-order covariant wave equation $\Box_g \sigma_\mu = Q_\mu + G_\mu + T_\mu + \kappa \ell_Q^2 \Box_g^2 \sigma_\mu + \mathcal{O}(\ell_Q^4)$ sourced by gravitational self-interaction, geometric coupling, matter, and quantum stiffness. The Pais–Uhlenbeck factorisation splits this into massless and Planck-mass sectors, admitting an exact Green's function solution. Einstein's field equations emerge as the equilibrium limit.

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1 From the Action to the Field Equations

Chapters 1–2 established the graviton field $\sigma_\mu = \partial_\mu S/(mc)$, the master metric

$$g_{\mu\nu}(\sigma) = T^\alpha{}_\mu T^\beta{}_\nu \left(M_{\alpha\beta} \circ \left[\eta_{\alpha\beta} - \sum_a \varepsilon_a \sigma^{(a)}_\alpha \sigma^{(a)}_\beta - \kappa \ell_Q^2 \partial_\alpha \sigma^\gamma \partial_\beta \sigma_\gamma \right] \right) \tag{1}$$

and the fundamental action

$$S[\sigma] = \int \mathrm{d}^4x \sqrt{-g(\sigma)} \left[\frac{R[g(\sigma)]}{16\pi G} + \frac{\hbar^2}{2M} g^{\mu\nu} \nabla_\mu \sigma^\alpha \nabla_\nu \sigma_\alpha + \mathcal{L}_{\text{matter}}(\psi, g(\sigma)) \right] \tag{2}$$

The metric is a composite field; $\sigma_\mu(x)$ is the true dynamical variable.

1.1 Variation with respect to σ_α

Define the Einstein error tensor $E^{\mu\nu} \equiv G^{\mu\nu} - (8\pi G/c^4) T^{\mu\nu}$. The metric variation induced by $\delta\sigma_\alpha$ is

$$\delta g_{\mu\nu} = \frac{\partial g_{\mu\nu}}{\partial \sigma_\alpha} \delta\sigma_\alpha = -(\sigma_\mu \delta\sigma_\nu + \sigma_\nu \delta\sigma_\mu) + (\text{Kerr–Schild and radiation terms}) \tag{3}$$

The total variation yields

$$\boxed{\frac{\hbar^2}{M} \nabla^2 \sigma^\alpha = \frac{1}{16\pi G} E^{\mu\nu} \frac{\partial g_{\mu\nu}}{\partial \sigma_\alpha}} \tag{4}$$

This is the σ-field equation in compact form. **This equation does not exist in general relativity.**

2 The Master Field Equation

2.1 The Einstein–wave operator identity

Lemma 2.1 (Einstein–wave operator identity)**.** *For the QGD metric (1), the contraction of the Einstein tensor with σ_ν satisfies*

$$G^{\mu\nu}\sigma_\nu = -\Box_g \sigma^\mu + Q^\mu(\sigma, \partial\sigma) + G^\mu(\sigma, \ell, H, q) + \tfrac{1}{2} T^{\mu\nu}\sigma_\nu \tag{5}$$

Proof. From $G^{\mu\nu} = R^{\mu\nu} - \frac{1}{2} g^{\mu\nu} R$. The Christoffel symbols of the σ-sector give $\Gamma^\lambda{}_{\mu\nu}|_\sigma = g^{\lambda\rho}[\sigma_\rho \partial_{(\mu}\sigma_{\nu)} - \sigma_{(\mu}\partial_{\nu)}\sigma_\rho]$. Computing $R^{\mu\nu}\sigma_\nu$ and subtracting $\frac{1}{2} g^{\mu\nu} R \sigma_\nu$ yields (5) after collecting the nonlinear and coupling terms into Q^μ and G^μ. □

2.2 The complete master field equation

Evaluating the right-hand side of (4) using the identity (5) and including the quantum stiffness from the σ-kinetic term:

$$\boxed{\Box_g \sigma_\mu = Q_\mu(\sigma, \partial\sigma) + G_\mu(\sigma, \ell, H, q) + T_\mu + \kappa \ell_Q^2 \Box_g^2 \sigma_\mu + \mathcal{O}(\ell_Q^4)} \tag{6}$$

where $\Box_g = g^{\alpha\beta} \nabla_\alpha \nabla_\beta$ is the covariant d'Alembertian and $\ell_Q = \sqrt{G\hbar^2/c^4}$. The four terms:

(1) Quantum self-interaction Q_μ:

$$Q_\mu(\sigma, \partial\sigma) = \sigma_\mu(\nabla_\alpha\sigma_\beta\nabla^\alpha\sigma^\beta) + (\nabla_\mu\sigma_\alpha)(\sigma_\beta\nabla^\beta\sigma^\alpha) \tag{7}$$

Gravity gravitates. The graviton field's own energy curves spacetime. Analogous to gluon self-coupling $Q \sim A \cdot F$ in Yang–Mills.

(2) Geometric coupling G_μ:

$$G_\mu = \sum_A \Big[H_A(\ell^{(A)}\cdot\nabla)^2\sigma_\mu + (\nabla\sigma)\cdot\nabla(H_A\ell\ell)\Big] + \nabla_\alpha(q^{\alpha\beta}\nabla_\beta\sigma_\mu) \tag{8}$$

Strong-field geometry modifies graviton propagation. Vanishes in weak field.

(3) Matter source T_μ:

$$T_\mu = \tfrac{1}{2}T^{\mu\nu}\sigma_\nu \tag{9}$$

Universal coupling: every form of energy with $T^{\mu\nu} \neq 0$ sources graviton waves.

(4) Quantum stiffness $\kappa\ell_Q^2\Box_g^2\sigma_\mu$: The fourth-order term from the σ-kinetic action. Negligible at macroscopic scales ($\ell_Q \sim 10^{-35}$ m); resolves singularities at the Planck scale.

2.3 Comparison with gauge forces

Force	Field	Equation	Self-coupling
Electromagnetism	A_μ	$\Box A_\mu = J_\mu$	None
Yang–Mills	A_μ^a	$D_\mu F^{\mu\nu} = j^\nu$	$Q \sim A \cdot F$
Gravity (QGD)	σ_μ	$\Box_g\sigma_\mu = Q_\mu + G_\mu + T_\mu + \kappa\ell_Q^2\Box_g^2\sigma_\mu$	$Q \sim \sigma(\partial\sigma)^2$

3 Wave Propagation Regimes

Linear ($|\sigma| \ll 1$): $Q_\mu \approx 0$, $G_\mu = 0$, $\Box_g \to \Box_\eta$: $\Box_\eta\sigma_\mu = T_\mu + \kappa\ell_Q^2\Box_\eta^2\sigma_\mu$.

Quasilinear (moderate field): $\Box_g\sigma_\mu = Q_\mu + T_\mu + \kappa\ell_Q^2\Box_g^2\sigma_\mu$.

Nonlinear (strong field): Full equation (6) with all terms.

4 Gravitational Waves

4.1 Vacuum wave equation

Far from sources ($Q_\mu, G_\mu, T_\mu \to 0$), the master equation reduces to

$$\Box_g\sigma_\mu + \kappa\ell_Q^2\Box_g^2\sigma_\mu = 0 \tag{10}$$

At astrophysical scales ($r \gg \ell_Q$) the stiffness term is negligible, giving the pure wave equation $\Box_g\sigma_\mu = 0$. Plane wave solutions $\sigma_\mu = \epsilon_\mu e^{ik_\nu x^\nu}$ yield:

$$g^{\alpha\beta}k_\alpha k_\beta = 0 \quad \text{(null propagation)}, \qquad k^\mu\epsilon_\mu = 0 \quad \text{(transversality)} \tag{11}$$

Gravitational waves propagate at speed c, are transverse, with two polarisation states.

4.2 Gravitational waves from binary sources

The two-body cross-term $h_{\mu\nu}^{\mathrm{GW}} = -2\sigma_\mu^{(1)}\sigma_\nu^{(2)}$ gives, in TT gauge for a circular binary:

$$\boxed{h_+(t) = \frac{G}{c^4 r_{\mathrm{obs}}} M_{\mathrm{tot}} \left(\frac{d}{2}\right)^2 \omega^2 \cos(2\omega t)} \tag{12}$$

matching the standard quadrupole formula, derived from σ-field superposition.

4.3 LIGO interpretation

What LIGO detects is the oscillation of σ_μ: $\delta g_{\mu\nu} \sim -\sigma_\mu \delta\sigma_\nu - \delta\sigma_\mu \sigma_\nu$.

5 The Fourth-Order Equation and Pais–Uhlenbeck Factorisation

Rearranging (6) and collecting the quantum stiffness:

$$\boxed{(\Box_g - \kappa \ell_Q^2 \Box_g^2)\sigma_\mu = S_\mu} \tag{13}$$

where

$$S_\mu = \frac{8\pi G}{c^4} Q_\mu + \frac{8\pi G}{c^4} G_\mu + \frac{4\pi G}{c^2} T^{\mu\nu}\sigma_\nu \tag{14}$$

5.1 Factorisation

The left-hand side factors exactly:

$$\Box_g - \kappa \ell_Q^2 \Box_g^2 = \Box_g (1 - \kappa \ell_Q^2 \Box_g) \tag{15}$$

Setting $1 - \kappa \ell_Q^2 \Box_g = 0$ defines the Planck mass:

$$m_Q^2 = \frac{1}{\kappa \ell_Q^2} = \frac{M_{\mathrm{Pl}}^2 c^2}{\kappa \hbar^2}, \qquad m_Q = \frac{M_{\mathrm{Pl}} c}{\sqrt{\kappa}\,\hbar} \tag{16}$$

The factored form is:

$$\boxed{\Box_g (\Box_g - m_Q^2)\sigma_\mu = -m_Q^2 S_\mu} \tag{17}$$

Two modes:

1. **Massless:** $\Box_g \sigma_\mu^{(0)} = 0$ — classical gravitational waves at speed c.
2. **Massive:** $(\Box_g - m_Q^2)\sigma_\mu^{(m)} = 0$ — Planck-mass gravitons, exponentially damped on $\tau_Q = \hbar/(m_Q c^2) \sim 10^{-43}$ s. Unobservable at astrophysical scales.

6 Exact Solution via Green's Functions

6.1 Mode decomposition

Write $\sigma_\mu = \sigma_\mu^{(0)} + \sigma_\mu^{(m)}$ with

$$\Box_g \sigma_\mu^{(0)} = J_\mu, \qquad (\Box_g - m_Q^2)\sigma_\mu^{(m)} = -J_\mu \tag{18}$$

where $J_\mu = -\kappa \ell_Q^2 S_\mu$. Verification: $\Box_g(\Box_g - m_Q^2)(\sigma^{(0)} + \sigma^{(m)}) = (\Box_g - m_Q^2)J - \Box_g J = -m_Q^2 J = S_\mu$. ✓

6.2 Retarded Green's functions

$$\Box_g G_0(x,x') = \delta^{(4)}(x-x')/\sqrt{-g(x')} \quad \text{(massless)} \tag{19}$$

$$(\Box_g - m_Q^2) G_{m_Q}(x,x') = \delta^{(4)}(x-x')/\sqrt{-g(x')} \quad \text{(massive)} \tag{20}$$

In flat spacetime:

$$G_0(x-x') = \frac{\delta(t-t'-|\mathbf{x}-\mathbf{x}'|/c)}{4\pi|\mathbf{x}-\mathbf{x}'|} \tag{21}$$

$$G_{m_Q}(x-x') = \frac{\delta(t-t'-r/c)}{4\pi r} - \frac{m_Q}{4\pi}\frac{J_1(m_Q\sqrt{c^2(t-t')^2-r^2})}{\sqrt{c^2(t-t')^2-r^2}}\,\Theta(c(t-t')-r) \tag{22}$$

6.3 The exact integral solution

Theorem 6.1 (Exact integral solution)**.** *The general solution to* (13) *on a globally hyperbolic spacetime is:*

$$\boxed{\sigma_\mu(x) = \sigma_\mu^{\text{free}}(x) + \kappa\ell_Q^2 \int_{\mathcal{M}} \mathrm{d}^4x' \sqrt{-g(x')}[G_0(x,x') - G_{m_Q}(x,x')]S_\mu(x')} \tag{23}$$

The composite propagator $G_{\text{QGD}} = \kappa\ell_Q^2[G_0 - G_{m_Q}]$ separates into:

- Classical sector (G_0): light-cone propagation, $1/r$ potential, reproduces GR.
- Quantum sector ($-G_{m_Q}$): Yukawa-suppressed $e^{-m_Q r}/r$, decay length $\sim \ell_Q$, resolves singularities.

No superluminal propagation: both propagators are retarded; the massive one is strictly more localised.

6.4 Iterative solution: Born series

Since S_μ depends on σ through Q_μ and G_μ, Eq. (23) is an integral equation solved iteratively. Each iteration adds corrections of order $\epsilon^2 \sim (GM/rc^2)^2$.

7 Energy-Momentum of the Graviton Field

Contracting (6) with σ^μ:

$$T_\sigma^{\alpha\beta} = \nabla^\alpha\sigma_\mu\nabla^\beta\sigma^\mu - \tfrac{1}{2}g^{\alpha\beta}(\nabla\sigma)^2 + \kappa\ell_Q^2\text{-corrections} \tag{24}$$

This is a true tensor, positive-definite, locally conserved. For Schwarzschild: $\rho_{\text{grav}}(r) = GM/(4c^2r^3)$.

8 Equivalence to Einstein's Field Equations

Theorem 8.1 (QGD $\Rightarrow$ Einstein)**.** *Any solution of* $\Box_g\sigma_\mu = Q_\mu + G_\mu + T_\mu + \kappa\ell_Q^2\Box_g^2\sigma_\mu$ *generates a metric satisfying* $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$.

Proof. From the identity (5) and the field equation with $T_\mu = \frac{1}{2}T^{\mu\nu}\sigma_\nu$:

$$G^{\mu\nu}\sigma_\nu = -(Q^\mu + G^\mu + T^\mu) + Q^\mu + G^\mu + \tfrac{1}{2}T^{\mu\nu}\sigma_\nu = 0 \tag{25}$$

Combined with analogous projections from the companion field equations for H_A, $\ell_\mu^{(A)}$, $q_{\mu\nu}$, the error tensor $E^{\mu\nu} = G^{\mu\nu} - (8\pi G/c^4)T^{\mu\nu}$ vanishes identically. $\square$

Theorem 8.2 (Einstein $\Rightarrow$ QGD). *Any solution of $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ admits a Debney–Kerr–Schild decomposition into fields satisfying the QGD system.*

Corollary 8.3 (Mathematical equivalence). *QGD and GR have a one-to-one correspondence of classical solutions:*

$$\nabla^2\sigma = 0 \iff G_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu} \tag{26}$$

The distinction: GR constrains the metric directly; QGD evolves σ_μ via a wave equation and the metric emerges.

9 Ghost-Freedom and Stability

Theorem 9.1 (Ghost-freedom). *The QGD quantum stiffness is ghost-free for all modes with $m < M_{\mathrm{Pl}}$.*

Proof. Dispersion relation $\omega^2 = k^2c^2(1 + \kappa\ell_Q^2k^2c^2 + \mathcal{O}(\ell_Q^4))$ has positive correction for all k. Ghosts appear only at $m > 1/(\sqrt{\kappa}\ell_Q) = M_{\mathrm{Pl}}c/(\sqrt{\kappa}\hbar)$. □

10 The Post-Newtonian Hierarchy

Expanding $\sigma_\mu = \sum_{k=1}^{\infty}\epsilon^k\sigma_\mu^{(k)}$:

$$\Box_\eta\sigma_\mu^{(k)} = Q_\mu^{(k)}[\sigma^{(<k)}] + T_\mu^{(k)} + \kappa\ell_Q^2\Box_\eta^2\sigma_\mu^{(k)} \tag{27}$$

Linear at each order, sourced by lower orders. Including quantum corrections via $(\hbar, \epsilon)$ double expansion:

$$\Box_g\sigma_\mu^{(n,k)} = S_\mu^{(n,k)}[\sigma^{(n'<n,k'<k)}] + \kappa\ell_Q^2\Box_g^2\sigma_\mu^{(n,k)} \tag{28}$$

11 Summary

The complete QGD field dynamics:

Equation	Content
$\Box_g\sigma_\mu = Q_\mu + G_\mu + T_\mu + \kappa\ell_Q^2\Box_g^2\sigma_\mu$	Master field equation
$\Box_g(\Box_g - m_Q^2)\sigma_\mu = S_\mu$	Pais–Uhlenbeck form
$\sigma_\mu = \sigma^{\mathrm{free}} + \kappa\ell_Q^2\int(G_0 - G_{m_Q})S_\mu$	Green's function solution
$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$	Einstein at equilibrium